COMP1140 S2 2023
Assignment 2

Project: Database design of Numberone Pizza

Logical Database Design

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Part 1: Reflection on Assignment 1 and Assignment 2

This section presents necessary discussion to point out the differences between my submitted EER for assignment 2 and assignment 3.

In my initial submission for assignment 2, there were several areas that needed improvement. There included inconsistencies in attribute names throughout the document, the missing of foreign key for the Ingredient entity as well as missing relationship between the 'DriverStaff' and 'DriverShift' entities.

For the assignment 3, I have changed the name of some certain attributes in the Data Dictionary section to ensure consistency. Additionally, I have added new attributes ('DateIssued' and 'DateSupplied') into the IngredientOrder entity to enhance data tracking. I also aligned and formatted the table to display all the information. During the Relational Mapping Process, I added a foreign key ('SuppierID') from the Supplier into the IngredientOrder entity.

In the following discussions, ***the provided solution EER*** will be used in all the discussions.

Part 2: Requirements

Data Requirements

Order Processing:

An order can be placed over the phone or instore and be picked up or delivered. When an order is created, all information is recorded in the system.

Order Data:

To create the order in the database, it is necessary to gather some essential information about the order. This information includes the unique order number, time when order was initiated, information of the staff responsible for taking order, list of items ordered by customer (containing items name, quantities and individual prices), payment information (incorporate the total amount due and payment method), description as well as type of order itself. An order can be either phone order or walking order. A walk-in order includes the customer's walk-in time. For a phone order, the pickup time is recorded and the details includes the time the call was answered, the time the call was terminated and further information regarding delivery or pickup. A Delivery phone order comprises the customer's address, the delivery and the driver information. Last but not least, a walk-in order includes pickup time.

Customer Data:

When a customer orders, their phone number is entered into the system along with the id of the staff taking the order. If the customer has previously ordered, their name and address are display on the screen. The customer is then prompted to verify and provide their name and address, and then the order is taken. If the customer is new or the name and address given do not correspond with what is recorded in the computer, a new customer record is created, and the order is processed.

For each customer is created in database, their details entail a unique number to identity each customer, their first and last name, contact number, status, list of items previously ordered by the customer.

By collecting these customer's information, the Pizza Ordering System can improve customer information management, makes order processing more effectual and provide personalize service to enhance overall customer satisfaction.

Employee Data:

The employee entity includes information about in-store staff who work in the shop are paid hourly and delivery staff who carry out deliveries and are paid by number of deliveries.

Each employee's basic details, payment data and work hours is recorded.

Employee's data comprise an identify number, employee's first and second name, address, phone number, tax file number, bank detail (including bank code, bank name and account number, payment period start date, payment period end date), status and description.

Employee's payment data contain an id of each payment, payment record (including gross payment, tax withheld, total amount paid, payment date and bank details of the employee), and payment rate (only for in-store staff)

Shift data include the start time and data of each shift, as well as the ending time and data of each shift, shift type (in-store or delivery). For delivery staff, It also features the number of deliveries in the shift, order details (including delivery address and delivery time) while it contains the total number of work hours for an in-store staff.

Menu Items, Ingredients, Ingredient Order and Suppliers:

Menu Items data consist of all the details of available option for customer. These are defined by an identification number for each pizza option, name and size of pizza, the current price of each pizza, the quantity along with the identification numbers of all ingredients used to make the pizza.

Ingredients of each pizza include ingredient's code, name, type, description, current stock level, suggested stock level, reordered levels show when should restock the ingredient, total amount of ingredient, status, description, quantity and price of all ingredients data and supplier data (entail the supplier's name, phone number and email address)

Ingredient Order data contains crucial information for each ingredient order, including the order number, total amount of ingredients, ingredient code, status and description of each item.

Supplier data includes supplier's name for easy identifiable, supplier's contact number for order and delivery purpose, address and email for additional communicate channel.

Transaction Requirements

Data Manipulation

- Insert, Update and Delete existing Order.
- Insert, Update and Delete existing Customer.
- Insert, Update and Delete existing Customer Status.
- Insert, Update and Delete existing Ingredients.
- Insert, Update and Delete existing Ingredient Status.
- Insert, Update and Delete existing Menu Items.
- Insert, Update existing Employee.
- Insert, Update and Delete existing Employee Status.
- Insert, Update and Delete existing Supplier.
- Insert, Update and Delete existing Payment.

Queries: Search an item/ingredient/customer/employee based on a range of standard including keyword, name, type, code, id, range of price, ingredient, quantity, etc…

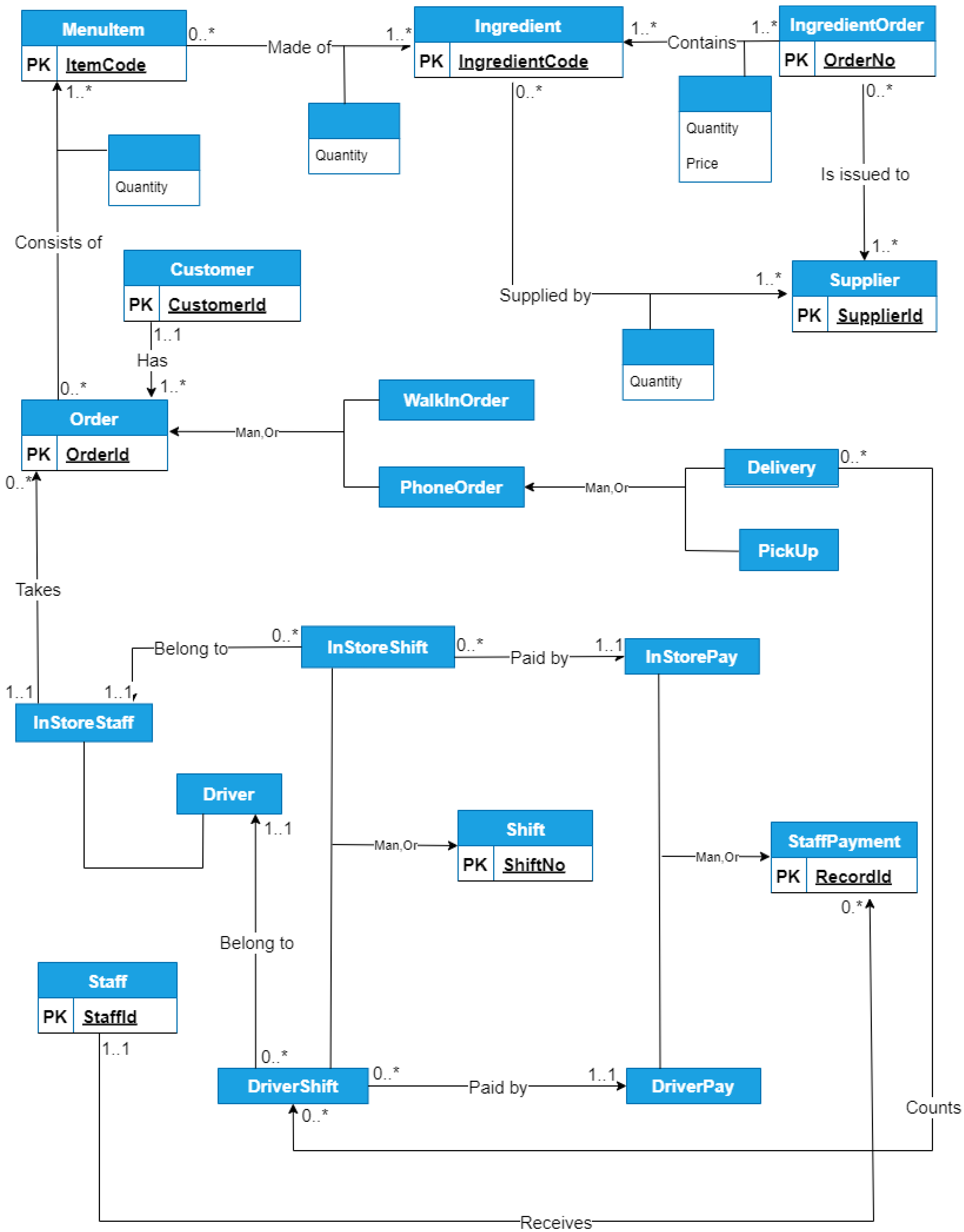
- Search a customer based on customer number.
- Search a customer based on customer’s phone number.
- Search a payment based on an employee number, on a particular pay date.
- Search a delivery bases on a customer order number.
- Search order’s type based on customer order number.
- Search an employee based on the name/id/shift/status.
- Search the items based on range of price.
- Search an item based on a code, on name.
- List ingredients and quantity used for menu items.
- List all the available items in the menu.
- Report of ingredient levels for current period and suggested stock levels.

Business Rules

- The amount of each ingredient remaining must be updated every time some is used.
- The results of the weekly stock count must be input into database.
- When an ingredients stock level decreases below its recorder level an order for the ingredient must be placed.
- An employee can only be either an in-store worker or delivery driver.
- Employees cannot delete data from the database.
- An employee can only be either an in-store worker or a delivery driver.
- An employees' status can only be either full time or part time.
- Payment can only be added by accounting staff.
- An orders payment method can only have one of the following values:
    - Credit card
    - Debit card
    - Cash
- An order's type can only be either:
    - Pick up
    - Delivery
- If an order is paid for using card, the approval number must be stored in the order's paymentApprovalNo

Part 3: EER Model with Data Dictionary

EER Model



Order	
PK	OrderID
	OrderDateTime
	OrderType
	TotalAmountDue
	PaymentMethod
	PaymentApprovalNo
	Status

Staff	
PK	StaffID
AK	TaxFileNo
	Name
	FirstName
	Surname
	Address
	Street
	City
	Postcode
	Phone
	BankAc
	AcName
	BSB
	AccNo
	Status
	Descriptiom

DriverPay
PaidDeliveryRate
DeliveriesPaid

InStorePay
PaidHourlyRate
HoursPaid

PhoneOrder
PhoneEndTime
PhoneStartTime

Customer	
PK	CustomerID
	Name
	FirstName
	Surname
	Address
	Street
	City
	Postcode
	Phone

Ingredient	
PK	IngredientCode
	Name
	Type
	CurrentStockLevel
	SuggestedStockLevel
	ReorderedLevel
	Description

StaffPayment	
PK	RecordID
	GrossPay
	TaxWithheld
	TotalAmountPaid
	PaymentDate
	PayPeriodStartDate
	PayPeriodEndDate

Driver
DriverLicNo
RatePerDelivery

InStoreStaff
HourlyRate

WalkInOrder
WalkInTime

MenuItem	
PK	ItemCode
	Name
	Size
	Price
	Description

IngredientOrder	
PK	IngredientOrderNo
	Total
	Description
	Status

Supplier	
PK	SupllierID
	Name
	Phone
	Address
	Street
	City
	Postcode
	Email

Shift	
PK	ShiftNo
	StartDateTime
	EndDateTime

InStoreShift
WorkHourNum

DriverShift
DeliveriesNum

Delivery
DeliveryAddress
DeliveryTime

PickUp
PickUpTime

Data Dictionary

Entity Type:

Entity	**Description**	**Aliases**	**Occurrences**
Order	Describing orders customers have made	Pizza Order	When an order is made by a customer
Customer	Providing information about the customer	Client	When customer places an order
			When a new customer is created in the Database
			When customer’s information is updated
Staff	Providing information about employees	Staff Information	When employee takes the order
			When employee’s information is updated
			When employee is created in the Database
StaffPayment	Providing payment details of employees	Staff Compensation	When employee gets paid
Shift	Recording start and end times of each shift	Shift Details	When employee starts or ends the working shift
MenuItem	Listing all items currently on sales	Menu Option	When a new dish is added to menu
Ingredient	Listing all the ingredients used in pizzas		When a new ingredient is added
IngredientOrder	Displaying an ingredient order	Ingredient Purchase	When restaurant purchase an ingredient order
Supplier	Providing information about suppliers	Provider	When supplier’s information is updated

PhoneOrder	The order customer makes through phone call	Phone Call	When an order is placed through phone call
WalkInOrder	The order customer makes by walk in the store	Face to Face	When customer makes an order in the store
Delivery	Delivery details (including time and address)	Drop off	When customer want an order to be handed over
PickUp	Including pick up time	In-store Pick up	When customer makes the order on the phone
InStoreShift	Recording number of working hours for in-store staff	Retail Shift	When in-store staff starts or ends the working shift
DriverShift	Tracking number of deliveries completed by driver	Delivery Shift	When delivery staff successfully delivery an order
DriverStaff	The staff who responsible for delivery order to customer	Delivery Staff	When driver deliveries an order
InStoreStaff	The staff who responsible for taking order inside the restaurant	Retail Staff	When an ordered is placed by the in-store staff
DriverPay	Payment for driver	Driver Payment	When a driver receives salary
InStorePay	Payment for in-store staff	Retail Staff Payment	When an in-store staff receives salary

Relationships Type

Entity Name	Multiplicity	Relationship	Multiplicity	Entity Name
Order	0..*	Consists of	1..*	MenuItem
	(Man,Or)	Generalisation	(Man,Or)	WalkInOrder
	(Man,Or)	Generalisation	(Man,Or)	PhoneOrder
Customer	1..1	Has	1..*	Order
PhoneOrder	(Man,Or)	Specialization	(Man,Or)	Delivery
	(Man,Or)	Specialisation	(Man,Or)	PickUp
MenuItem	0..*	Made of	1..*	Ingredient
Staff	1..1	Receives	0..*	StaffPayment
	(Man,Or)	Generalisation	(Man,Or)	DriverStaff
	(Man,Or)	Generalisation	(Man,Or)	InstoreStaff
InStoreStaff	1..1	Takes	0..*	Order
StaffPayment	(Man,Or)	Generalisation	(Man,Or)	DriverPay
	(Man,Or)	Generalisation	(Man,Or)	InStorePay
IngredientOrder	0..*	Is issued to	1..*	Supplier
IngredientOrder	1..*	Contains	1..*	Ingredient
Ingredient	0..*	Supplied by	1..*	Supplier
InStoreShift	0..*	Belongs to	1..1	InStoreStaff
DriverShift	0..*	Belongs to	1..1	DriverStaff
Delivery	0..*	Counts	0..*	DriverShift
Shift	(Man,Or)	Generalisation	(Man,Or)	DriverShift
	(Man,Or)	Generalisation	(Man,Or)	InStoreShift
InStoreShift	0..*	Paid by	1..1	InStorePay

DriverShift	0..*	Paid by	1..1	DriverPay

Attributes Type:

Entity	Attributes	Description	Data Type & Length	Nulls	Multi-valued	Derived	Default
Order	OrderID	A unique order identifier	Char(10)	N	N	N	
	OrderDateTime	The date & time the order is made.	Datetime	Y	N	N	
	OrderType	Walking or Phone order	Varchar(10)	Y	N	N	
	TotalAmountDue	Total money for the order	Float	Y	N	Y	
	PaymentMethod	Total 3 kinds of payment	Varchar(10)	N	N	N	
	PaymentApproval No	Payment Approval No	Char(10)	N	N	N	
	Status	Current state of the order, including executed or not	Varchar(50)	Y	N	N	
Customer	CustomerID	Unique number to identify each customer	Varchar(10)	N	N	N	
	Name						
	FirstName	Customer’s first name	Varchar(20)	Y	N	N	
	Surname	Customer’s last name	Varchar(10)	Y	N	N	
	Phone	Customer’s contact number	Varchar(14)	Y	Y	N	
	Address	Address of the Customer		Y	N	N	
	Street	Name of the street where the customer live	Varchar(20)	Y	N	N	

	City	Name of the city where the customer live	Varchar(10)	Y	N	N	
	Postcode	Postcode number whe the customer live	Varchar(4)	Y	N	N	
Staff	StaffID	Unique staff identifier	Varchar(10)	N	N	N	
	Name						
	FirstName	Staff’s first name	Varchar(20)	Y	N	N	
	Surname	Staff’s last name	Varchar(10)	Y	N	N	
	Address	Staff’s postal address.	Varchar(50)	Y	N	N	
	Street	Name of the street where the staff live	Varchar(20)	Y	N	N	
	City	Name of the city where the staff live	Varchar(10)	Y	N	N	
	Postcode	Postcode number where the staff live	Varchar(4)	Y	N	N	
	Phone	Staff’s contact number	Varchar(20)	Y	Y	N	
	TaxFileNum	Staff’s tax file number	Char(9)	N	N	N	
	BankDetails						
	AcName	Staff’s bank account name	Char(6)	Y	N	N	

	BSB	BSB number of the bank	Varchar(15)	Y	N	N	
	AcNo	Staff's bank account number	Varchar(15)	Y	N	N	
	Status	Status of the staff	Varchar(50)	Y	N	N	
	Description	Additional information about the staff	Text	Y	N	N	
StaffPayment	RecordID	Unique payment record identifier	Varchar(20)	N	N	N	
	GrossPay	Staff's gross salary	Float	Y	N	N	
	TaxWithheld	Tax withheld	Float	Y	N	N	
	TotalAmountPaid	Total payment made to staff	Float	Y	N	N	
	PaymentDate	Payment date	Date	Y	N	N	
	PayPeriodStartDate	Payment period start date	Date	Y	N	N	
	PayPeriodEndDate	Payment period end date	Date	Y	N	N	
Shift	ShiftID	Unique number to identify the shift	Varchar(10)	N	N	N	
	StartDateTime	The start date and time of each shift	Datetime	Y	N	N	
	EndDateTime	The end date of each shift	Date	Y	N	N	
MenuItem	ItemCode	An unique number for each pizza option.	Varchar(10)	N	N	N	

	Name	The name of each pizza	Varchar(10)	Y	N	N	
	Size	Providing different choices of size	Double	Y	N	N	
	Price	The current price of each pizza	Float	Y	N	N	
	Description	Additional information about the item	Text	Y	N	N	
Ingredient	IngredientCode	A unique number to identify each ingredient	Varchar(10)	N	N	N	
	Name	The name of ingredient	Varchar(50)	Y	N	N	
	Type	The type of ingredient for management purpose	Varchar(20)	Y	N	N	
	Description	Description about the ingredient	Text	Y	N	N	
	CurrentStockLevel	The stock level of ingredient at current period	Varchar(20)	Y	N	N	
	SuggestedStockLevel	The recommended stock level of ingredient	Varchar(20)	Y	N	N	
	ReorderedLevel	The reordered level show when should restock the ingredient	Varchar(20)	Y	N	N	
IngredientOrder	IngredientOrderNo	An unique number for each ingredient order	Varchar(20)	N	N	N	
	TotalAmount	The total amount of ingredient ordered	Decimal(10, 2)	Y	N	N	
	DateIssued	The date when the order is issued	Datetime	Y	N	N	
	DateSupplied	The date when the order is supplied	Datetime	Y	N	N	

	Description	Description of the ingredient order	Text	Y	N	N	
	Status	Status of the ingredient order	Varchar(50)	Y	N	N	
Supplier	SupplierID	Unique number to identify each supplier	Varchar(10)	N	N	N	
	Name	The supplier's name, for easily identifiable	Varchar(50)	N	N	N	
	Phone	The supplier's contact number	Varchar(20)	Y	N	N	
	Address	The supplier address					
	Street	Name of the street where the supplier is located	Varchar(20)	Y	N	N	
	City	Name of the city where the supplier is located	Varchar(10)	Y	N	N	
	Postcode	Postcode number where the supplier is located	Varchar(4)	Y	N	N	
	Email	The supplier's email for additional communicate channel	Varchar(30)	Y	N	N	
PhoneOrder	PhoneStartTime	The time when the call is answered	Datetime	Y	N	N	
	PhoneEndTime	The time when the call is ended	Datetime	Y	N	N	
WalkInOrder	WalkInTime	The time when customer walks in	Datetime	Y	N	N	
Delivery	DeliveryAddress	Delivery address (for delivery order only)	Varchar(100)	Y	N	N	
	DeliveryTime	Delivery time (for delivery order only)	Datetime	Y	N	N	

PickUp	PickUpTime	The time when customer pickup an order	Datetime	Y	N	N	
DriverStaff	DriveLicenseNo	Driver’s License number	Varchar(15)	Y	N	N	
	RatePerDelivery	Payment rate for each delivery	Decimal(10, 2)	Y	N	N	
InStoreStaff	HourlyRate	Payment rate for in-store staff	Decimal(10, 2)	Y	N	N	
DriverShift	DeliveriesNum	Number of deliveries in the shift (for driver)	Int	Y	N	N	
InStoreShift	WorkHourNum	Number of working hours (for in-store staff)	Int	Y	N	N	
DrivePay	DeliveriesPaid	The total deliveries for which a driver is paid	Int	Y	N	N	
	PaidDeliveryRate	The delivery rate which is paid for driver	Decimal(10, 2)	Y	N	N	
InStorePay	HoursPaid	The total hours for which an in-store staff is paid	Int	Y	N	N	
	PaidHourlyRate	The hourly rate which is paid for in-store staff	Decimal(10, 2)	Y	N	N	

Part 4: Mapping the EER to Relational Model

Using the mapping rules, got the following relations for all entities in EER.

Note: For Order, map Orders into sup+individual sub classes

Order (OrderID, OrderDateTime, TotalAmountDue, PaymentMethod, PaymentApprovalNo, Status, CustomerId, StaffId)

PRIMARY KEY OrderID

FOREIGN KEY CustomerID REFERENCES Customer(CustomerID) ON UPDATE CASCADE, ON DELETE CASCADE

FOREIGN KEY StaffID REFERENCES InstoreStaff(StaffID) ON UPDATE CASCADE, ON DELETE CASCADE

WalkInOrder (OrderID, WalkInTime)
PRIMARY KEY OrderID
FOREIGN KEY OrdeID REFERENCES Order (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE

PhoneOrder (OrderID, PhoneStartTime,PhoneEndTime)
PRIMARY KEY OrderID
FOREIGN KEY OrderID REFERENCES Order (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE

Delivery (OrderID, DeliveryAddress,DeliveryTime,ShiftID)
PRIMARY KEY OrderID
FOREIGN KEY OrderID REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE
FOREIGN KEY ShiftID REFERENCES DriverShift (ShiftID) ON UPDATE CASCADE, ON DELETE CASCADE

Pickup (OrderID, PickupTime)
PRIMARY KEY OrderID
FOREIGN KEY OrderId REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE

Customer (CustomerID, FirstName, LastName, Street, City, Postcode, Phone)
PRIMARY KEY CustomerID

DriverStaff (StaffID, FirstName, LastName, Phone, Street, City, Postcode, TaxFileNo, AcName, BSB, AccNo, Status, Description, DriverLicNo, RatePerDelivery)

PRIMARY KEY StaffID

AlTERNATE KEY TaxFileNo, DriverLicNo

InStoreStaff (StaffID, FirstName, LastName, Phone, Street, City, Postcode, TaxFileNo, AcName, BSB, AccNo, HourlyRate, Status, Description)

PRIMARY KEY StaffID

AlTERNATE KEY TaxFileNo

MenuItem (ItemCode, Name, Size, Description, Price)

PRIMARY KEY ItemCode

QMenuOrder (OrderID, ItemCode, Quantity)

PRIMARY KEY OrderID, ItemCode

FOREIGN KEY OrderID REFERENCES Order (OrderID) ON UPDATE CASCADE ON DELETE NO ACTION FOREIGN KEY ItemCode REFERENCES MenuItem (ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION

Ingredient (IngredientCode, Name, Type, Description, CurrentStockLevel, SuggestedStockLevel, ReorderLevel)

PRIMARY KEY IngredientCode

QMenuIngredient (ItemCode, IngredientCode, Quantity)

PRIMARY KEY ItemCode, IngredientCode

FOREIGN KEY ItemCode REFERENCES MenuItem (ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION

FOREIGN KEY IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION

IngredientOrder (IngredientOrderNo, TotalAmount, DateIssued, DateSupplied, Status, Description, SupplierID)

PRIMARY KEY IngredientOrderNo

Foreign Key SupplierID REFERENCES Supplier (SupplierID) ON UPDATE CASCADE ON DELETE NO ACTION

QIngredientIngredientOrder (IngredientCode, IngredientOrderNo, Price, Quantity)

PRIMARY KEY IngredientCode, IngredientOrderNo

FOREIGN KEY IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION

FOREIGN KEY IngredientOrderNo REFERENCES IngredientOrder (IngredientOrderNo) ON UPDATE CASCADE ON DELETE NO ACTION

Supplier (SupplierID, Name, Phone, Street, City, Postcode, Email)

PRIMARY KEY SupplierID

QSupplierIngredient (SupplierID, IngredientCode, Quantity)

PRIMARY KEY SupplierID

FOREIGN KEY IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION

DriverShift (ShiftID, StaffID, RecordID, StartDateTime, EndDateTime, DeliveriesNum)

PRIMARY KEY ShiftID

FOREIGN KEY StaffID REFERENCES Driver (StaffID) ON UPDATE CASCADE ON DELETE NO ACTION FOREIGN KEY RecordID REFERENCES DriverPay (RecordID) ON UPDATE CASCADE ON DELETE NO ACTION

InStoreShift (ShiftID, StaffID, RecordID, StartDateTime, EndDateTime, WorkHourNum)

PRIMARY KEY ShifID

FOREIGN KEY StaffID REFERENCES InStoreStaff (StaffID) ON UPDATE CASCADE ON DELETE NO ACTION

FOREIGN KEY RecordID REFERENCES InStorePay (RecordID) ON UPDATE CASCADE ON DELETE NO ACTION

DriverPay (RecordID, GrossPay, TaxWithheld, TotalAmountPaid, PaymentDate, PayPeriodStartDate, PayPeriodEndDate, PaidDeliveryRate, DeliveriesPay)

PRIMARY KEY RecordID

InStorePay (RecordID, GrossPay, TaxWithheld, TotalAmountPaid, PaymentDate, PayPeriodStartDate, PayPeriodEndDate, PaidHourlyRate, HoursPaid)

PRIMARY KEY RecordID

FOREIGN KEY OrderID REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE ON DELETE NO ACTION

Part 5: Normalising the Scheme up to BCNF

According to the definitions of 1NF, 2NF, 3NF and BCNF, it is identified that relations x1, x2, ..xn are all in BCNF, since all the attributes are atomic, and there exists only one function dependency in each table, and the left side of the FD is a PK.
But the following relations are not in BCNF. They are normalized as below.

Orders (OrderID, OrderDateTime, TotalAmountDue, PaymentMethod, PaymentApprovalNo, Status, CustomerID, StaffID)
Primary Key OrderID
Foreign Key CustomerID REFERENCES Customer(CustomerID) ON UPDATE CASCADE, ON DELETE CASCADE
Foreign Key StaffID REFERENCES InstoreStaff(StaffID) ON UPDATE CASCADE, ON DELETE CASCADE

FD1: OrderID -> OrderDateTime, TotalAmountDue, PaymentMethod, PaymentApprovalNo, Status
FD2: PaymentApprovalNo -> PaymentMethod, TotalAmountDue

Therefore, there exists transitive dependency, so the relation is in 2nd but not 3rd NF.
The normalization process:
OrderOnly ((OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerId, StaffID)
Primary Key OrderNo
Foreign Key CustomerID REFERENCES Customer(CustomerID) ON UPDATE CASCADE, ON DELETE CASCADE
Foreign Key StaffID REFERENCES InstoreStaff(StaffID) ON UPDATE CASCADE, ON DELETE CASCADE
Foreign Key PaymentApprovalNo REFERENCES OrderPayRecord (PaymentApprovalNo) ON UPDATE CASCADE, ON DELETE CASCADE
OrderPayRecord (PaymentApprovalNo, PaymentMethod, TotalAmountDue)
Primary Key PaymentApprovalNo
Note: Since the Orders is split into 2 tables, the relationships of original Orders to all the other tables should be reconsidered now – use OrderOnly to connect to other tables.

Customer (CustomerID, FirstName, LastName, Street, City, Postcode, Phone)
Primary Key CustomerID
Customer is not in normalized form due to multi-valued attributes for Phone.
To get it into 1st Normal Form (1NF), by eliminating the multi-valued attributes, the Phone attribute is decomposed into separate table
After decomposition, we have two tables:
Customer (CustomerID, FirstName, LastName, Street, City, Postcode)
Primary Key CustomerID
CPhone (Phone, CustomerID)
Primary Key Phone

Foreign Key CustomerID REFERENCES Customer (CustomerID) ON UPDATE CASCADE ON DELETE CASCADE
Both tables are now in Boyce-Codd Normal Form (BCNF) without partial or transitive dependencies

DriverStaff (StaffID, FirstName, LastName, Phone, Street, City, Postcode, TaxFileNo, AcName, BSB, AccNo, Status, Description, DriverLicNo, RatePerDelivery)

FD1: StaffID → (FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AcName, AccNo, Status, Description, DriverLicNo, RatePerDelivery)

FD2: BSB → AcName [Transitive Dependency]

While DriverStaff is in 2nd Normal Form (2NF), it is not in 3rd Normal Form (3NF) due to the transitive dependency involving BSBS and BankName.
Decompose into two tables based on the dependencies.
DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, Status, Description, DriverLicNo, RatePerDelivery)
Primary Key StaffID
Alternate KeyTaxFileNo, LicenceNo
Foreign Key BSB REFERENCES BankDetails (BSB) ON UPDATE CASCADE ON DELETE CASCADE
Bank Details (BSB, AcName)
Primary Key BSB
Both tables are now in Boyce-Codd Normal Form (BCNF) without partial or transitive dependencies

InStoreStaff (StaffID, FirstName, LastName, Phone, Street, City, Postcode, TaxFileNo, BSB, AcName, AccNo, HourlyRate, Status, Description)
Normalization process is similar to DriverStaff. After decomposition, we have two tables:
InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB AccNo, HourlyRate, Status, Description)
Primary Key StaffID
Alternate Key TaxFileNo
Foreign Key BSB REFERENCES BankDetails (BSB) ON UPDATE CASCADE ON DELETE CASCADE
Bank Details (BSB, AcName)
Primary Key BSB
Both tables are now in Boyce-Codd Normal Form (BCNF) without partial or transitive dependencies

WalkInOrder (OrderId, WalkInTime)
Primary Key OrderID
Foreign Key OrdeID REFERENCES Order (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE

PhoneOrder (OrderId, PhoneStartTime,PhoneEndTime)
Primary Key OrderID
Foreign Key OrderID REFERENCES Order (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE

Delivery (OrderId, DeliveryAddress,DeliveryTime,ShiftID)
Primary Key OrderID
Foreign Key OrderID REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE ON DELETE CASCADE
Foreign Key ShiftID REFERENCES DriverShift (ShiftID) ON UPDATE NO ACTION ON DELETE CASCADE

Pickup (OrderId, PickupTime)
Primary Key OrderID
Foreign Key OrderId REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE ON DELETE CASCADE

MenuItem (ItemCode, Name, Size, Description, Price)
Primary Key ItemCode

Ingredient (IngredientCode, Name, Type, Description, CurrentStockLevel, SuggestedStockLevel, ReorderLevel)
Primary Key IngredientCode

QMenuOrder (OrderID, ItemCode, Quantity)
Primary Key OrderID, ItemCode
Foreign Key OrderID REFERENCES Order (OrderID) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key ItemCode REFERENCES MenuItem (ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION

QMenuIngredient (ItemCode, IngredientCode, Quantity)
Primary Key ItemCode, IngredientCode
Foreign Key ItemCode REFERENCES MenuItem (ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION

IngredientOrder (IngredientOrderNo, TotalAmount, DateIssued, DateSupplied, Status, Description, SupplierID)
Primary Key IngredientOrderNo
Foreign Key SupplierID REFERENCES Supplier (SupplierID) ON UPDATE CASCADE ON DELETE NO ACTION
QIngredientIngredientOrder(IngredientCode, IngredientOrderNo, Price, Quantity)
Primary Key IngredientCode, IngredientOrderNo

Foreign Key IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key IngredientOrderNo REFERENCES IngredientOrder (IngredientOrderNo) ON UPDATE CASCADE ON DELETE NO ACTION

Supplier (SupplierID, Name, Phone, Street, City, Postcode, Email)
Primary Key SupplierID

DriverShift (ShiftID, StaffID, RecordID, StartDateTime, EndDateTime, DeliveriesNum)
Primary Key ShiftID
Foreign Key StaffID REFERENCES Driver (StaffID) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key RecordID REFERENCES DriverPay (RecordID) ON UPDATE CASCADE ON DELETE NO ACTION

InStoreShift (ShiftID, StaffID, RecordID, StartDateTime, EndDateTime, WorkHourNum)
Primary Key ShifID
Foreign Key StaffID REFERENCES InStoreStaff (StaffID) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key RecordID REFERENCES InStorePay (RecordID) ON UPDATE CASCADE ON DELETE NO ACTION

DriverPay (RecordID, GrossPay, TaxWithheld, TotalAmountPaid, PaymentDate, PayPeriodStartDate, PayPeriodEndDate, PaidDeliveryRate, DeliveriesPay)
Primary Key RecordID

InStorePay (RecordID, GrossPay, TaxWithheld, TotalAmountPaid, PaymentDate, PayPeriodStartDate, PayPeriodEndDate, PaidHourlyRate, HoursPaid)
Primary Key RecordID
Foreign Key OrderID REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE ON DELETE NO ACTION

Finally, the list of all the normalized tables is:

OrderOnly ((OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerId, StaffID)
Primary Key OrderID
Foreign Key CustomerID REFERENCES Customer(CustomerID) ON UPDATE CASCADE, ON DELETE CASCADE
Foreign Key StaffID REFERENCES InstoreStaff(StaffID) ON UPDATE CASCADE, ON DELETE CASCADE
Foreign Key PaymentApprovalNo REFERENCES OrderPayRecord (PaymentApprovalNo) ON UPDATE CASCADE, ON DELETE CASCADE

OrderPayRecord (PaymentApprovalNo, PaymentMethod, TotalAmountDue)
Primary Key PaymentApprovalNo

Customer (CustomerID, FirstName, LastName, Street, City, Postcode)
Primary Key CustomerID

CPhone (Phone, CustomerID)
Primary Key Phone
FOREIGN KEY CustomerID REFERENCES Customer (CustomerID) ON UPDATE CASCADE ON DELETE CASCADE

DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, Status, Description, DriverLicNo, RatePerDelivery)
Primary Key StaffID
ALTERNATE KEY TaxFileNo, LicenceNo
FOREIGN KEY BSB REFERENCES BankDetails (BSB) ON UPDATE CASCADE ON DELETE CASCADE

Bank Details (BSB, AcName)
Primary Key BSB

InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB AccNo, HourlyRate, Status, Description)
Primary Key StaffID
Alternate Key TaxFileNo
Foreign Key BSB REFERENCES BankDetails (BSB) ON UPDATE CASCADE ON DELETE CASCADE

WalkInOrder (OrderId, WalkInTime)
Primary Key OrderID
Foreign Key OrderID REFERENCES Order (OrderID) ON UPDATE CASCADE ON DELETE NO ACTION

PhoneOrder (OrderId, PhoneStartTime,PhoneEndTime)
Primary Key OrderID
Foreign Key OrderID REFERENCES Order (OrderID) ON UPDATE CASCADE ON DELETE CASCADE

Delivery (OrderId, DeliveryAddress,DeliveryTime,ShiftID)
Primary Key OrderID
Foreign Key OrderID REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE ON DELETE NO ACTIOn
Foreign Key ShiftID REFERENCES DriverShift (ShiftID) ON UPDATE CASCADE ON DELETE CASCADE

Pickup (OrderId, PickupTime)
Primary Key OrderID
Foreign Key OrderId REFERENCES PhoneOrder (OrderID) ON UPDATE CASCADE, ON DELETE CASCADE

MenuItem (ItemCode, Name, Size, Description, Price)
Primary Key ItemCode

Ingredient (IngredientCode, Name, Type, Description, CurrentStockLevel, SuggestedStockLevel, ReorderLevel)
Primary Key IngredientCode

QMenuOrder (OrderID, ItemCode, Quantity)
Primary Key OrderID, ItemCode
Foreign Key OrderID REFERENCES OrderOnly (OrderID) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key ItemCode REFERENCES MenuItem (ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION

QMenuIngredient (ItemCode, IngredientCode, Quantity)
Primary Key ItemCode, IngredientCode
Foreign Key ItemCode REFERENCES MenuItem (ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION

IngredientOrder (IngredientOrderNo, TotalAmount, DateIssued, DateSupplied, Status, Description, SupplierID)
Primary Key IngredientOrderNo

Foreign Key SupplierID REFERENCES Supplier (SupplierID) ON UPDATE CASCADE ON DELETE NO ACTION

QIngredientIngredientOrder (IngredientCode, IngredientOrderNo, Price, Quantity)
Primary Key IngredientCode, IngredientOrderNo
Foreign Key IngredientCode REFERENCES Ingredient (IngredientCode) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key IngredientOrderNo REFERENCES IngredientOrder (IngredientOrderNo) ON UPDATE CASCADE ON DELETE NO ACTION

Supplier (SupplierID, Name, Phone, Street, City, Postcode, Email)
Primary Key SupplierID

DriverShift (ShiftID, StaffID, RecordID, StartDateTime, EndDateTime, DeliveriesNum)
Primary Key ShiftID
Foreign Key StaffID REFERENCES DriverStaff (StaffID) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key RecordID REFERENCES DriverPay (RecordID) ON UPDATE CASCADE ON DELETE NO ACTION

InStoreShift (ShiftID, StaffID, RecordID, StartDateTime, EndDateTime, WorkHourNum)
Primary Key ShifID
Foreign Key StaffID REFERENCES InStoreStaff (StaffID) ON UPDATE CASCADE ON DELETE NO ACTION
Foreign Key RecordID REFERENCES InStorePay (RecordID) ON UPDATE CASCADE ON DELETE NO ACTION

DriverPay (RecordID, GrossPay, TaxWithheld, TotalAmountPaid, PaymentDate, PayPeriodStartDate, PayPeriodEndDate, PaidDeliveryRate, DeliveriesPay)
Primary Key RecordID

InStorePay (RecordID, GrossPay, TaxWithheld, TotalAmountPaid, PaymentDate, PayPeriodStartDate, PayPeriodEndDate, PaidHourlyRate, HoursPaid)
Primary Key RecordID

Part 6: SQL Script

```
DROP TABLE QIngredientIngredientOrder
DROP TABLE QMenuIngredient
DROP TABLE QMenuOrder
DROP TABLE Pickup
DROP TABLE Delivery
DROP TABLE PhoneOrder
DROP TABLE WalkInOrder
DROP TABLE OrderOnly
DROP TABLE OrderPayRecord
DROP TABLE InStoreShift
DROP TABLE DriverShift
DROP TABLE InstoreStaff
DROP TABLE DriverStaff
DROP TABLE BankDetails
DROP TABLE InstorePay
DROP TABLE DriverPay
DROP TABLE CPhone
DROP TABLE Customer
DROP TABLE IngredientOrder
DROP TABLE Supplier
DROP TABLE Ingredient
DROP TABLE MenuItem
go

-- Create the MenuItem table
CREATE TABLE MenuItem (
    ItemCode VARCHAR(10) de PRIMARY KEY,
    Name VARCHAR(50),
    Size VARCHAR(20),
    Description TEXT,
    Price DECIMAL(10, 2)
);
go

-- Create the Ingredient table
CREATE TABLE Ingredient (
    IngredientCode VARCHAR(10) NOT NULL PRIMARY KEY,
    Name VARCHAR(50),
    Type VARCHAR(20),
    Description TEXT,
    CurrentStockLevel VARCHAR(20),
    SuggestedStockLevel VARCHAR(20),
    ReorderLevel VARCHAR(20)
);
go

-- Create the Supplier table
CREATE TABLE Supplier (
    SupplierID VARCHAR(10) NOT NULL PRIMARY KEY,
    Name VARCHAR(50),
    Phone VARCHAR(20),
    Street VARCHAR(20),
    City VARCHAR(10),
    Postcode VARCHAR(4),
    Email VARCHAR(30)
);
go
```

```sql
-- Create the IngredientOrder table
CREATE TABLE IngredientOrder (
    IngredientOrderNo VARCHAR(10) NOT NULL PRIMARY KEY,
    TotalAmount DECIMAL(10, 2),
    DateIssued Date,
    DateSupplied Date,
    Status VARCHAR(50),
    Description TEXT,
    SupplierID VARCHAR(10),
    FOREIGN Key (SupplierID) REFERENCES Supplier(SupplierID) ON UPDATE CASCADE ON DELETE NO ACTION
);
go

-- Create the Customer table
CREATE TABLE Customer (
    CustomerID VARCHAR(10) NOT NULL PRIMARY KEY,
    FirstName VARCHAR(20),
    LastName VARCHAR(10),
    Street VARCHAR(20),
    City VARCHAR(10),
    Postcode VARCHAR(4),

);
go

-- Create the CPhone table
CREATE TABLE CPhone (
    Phone VARCHAR(14) PRIMARY KEY,
    CustomerID VARCHAR(10),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID) ON UPDATE CASCADE ON DELETE CASCADE
);
go

-- Create the DriverPay table
CREATE TABLE DriverPay (
    RecordID VARCHAR(10) NOT NULL PRIMARY KEY,
    GrossPay DECIMAL(10, 2),
    TaxWithheld DECIMAL(10, 2),
    TotalAmountPaid DECIMAL(10, 2),
    PaymentDate DATE,
    PayPeriodStartDate DATE,
    PayPeriodEndDate DATE,
    PaidDeliveryRate DECIMAL(10, 2),
    DeliveriesPay DECIMAL(10, 2)
);
go

-- Create the InStorePay table
CREATE TABLE InStorePay (
    RecordID VARCHAR(10) NOT NULL PRIMARY KEY,
    GrossPay DECIMAL(10, 2),
    TaxWithheld DECIMAL(10, 2),
    TotalAmountPaid DECIMAL(10, 2),
    PaymentDate DATE,
    PayPeriodStartDate DATE,
    PayPeriodEndDate DATE,
       PaidHourlyRate DECIMAL(10,2),
       HoursPaid INT
);
go
```

```sql
-- Create the BankDetails table
CREATE TABLE BankDetails (
    BSB VARCHAR(15) PRIMARY KEY,
    AcName VARCHAR(25)
);
go

-- Create the DriverStaff table
CREATE TABLE DriverStaff (
    StaffID VARCHAR(10) NOT NULL PRIMARY KEY,
    FirstName VARCHAR(20),
    LastName VARCHAR(10),
    Street VARCHAR(20),
    City VARCHAR(10),
    Postcode VARCHAR(4),
    TaxFileNo CHAR(9) ,
    BSB VARCHAR(15),
    AccNo VARCHAR(15),
    Status VARCHAR(50) DEFAULT 'Active',
    Description VARCHAR(20) DEFAULT 'Full-time' ,
    DriverLicNo VARCHAR(15),
    RatePerDelivery DECIMAL(10, 2),
    UNIQUE (TaxFileNo, DriverLicNo),
    FOREIGN KEY (BSB) REFERENCES BankDetails(BSB) ON UPDATE CASCADE ON DELETE CASCADE
);
go

-- Create the InStoreStaff table
CREATE TABLE InStoreStaff (
    StaffID VARCHAR(10) NOT NULL PRIMARY KEY,
    FirstName VARCHAR(20),
    LastName VARCHAR(10),
    Street VARCHAR(20),
    City VARCHAR(10),
    Postcode VARCHAR(4),
    TaxFileNo CHAR(9),
    BSB VARCHAR(15),
    AccNo VARCHAR(15),
    HourlyRate DECIMAL(10, 2),
    Status VARCHAR(50) DEFAULT 'Active',
    Description VARCHAR(20) DEFAULT 'Full-time',
       UNIQUE (TaxFileNo),
    FOREIGN KEY (BSB) REFERENCES BankDetails(BSB) ON UPDATE CASCADE ON DELETE CASCADE
);
go

-- Create the DriverShift table
CREATE TABLE DriverShift (
    ShiftID VARCHAR(10) NOT NULL PRIMARY KEY,
    StaffID VARCHAR(10),
    RecordID VARCHAR(10),
    StartDateTime DATETIME,
    EndDateTime DATETIME,
    DeliveriesNum INT,
    FOREIGN KEY (StaffID) REFERENCES DriverStaff(StaffID) ON UPDATE CASCADE ON DELETE NO ACTION,
    FOREIGN KEY (RecordID) REFERENCES DriverPay(RecordID) ON UPDATE CASCADE ON DELETE NO ACTION
);
go

-- Create the InStoreShift table
```

```sql
CREATE TABLE InStoreShift (
    ShiftID VARCHAR(10) NOT NULL PRIMARY KEY,
    StaffID VARCHAR(10),
    RecordID VARCHAR(10),
    StartDateTime DATETIME,
    EndDateTime DATETIME,
    WorkHourNum INT,
    FOREIGN KEY (StaffID) REFERENCES InStoreStaff(StaffID) ON UPDATE CASCADE ON DELETE NO ACTION,
    FOREIGN KEY (RecordID) REFERENCES InStorePay(RecordID) ON UPDATE CASCADE ON DELETE NO ACTION
);
go

-- Create the OrderPayRecord table
CREATE TABLE OrderPayRecord (
    PaymentApprovalNo VARCHAR(10) PRIMARY KEY,
    PaymentMethod VARCHAR(255),
    TotalAmountDue DECIMAL(10, 2)
);
go

-- Create the OrderOnly table
CREATE TABLE OrderOnly (
    OrderID VARCHAR(10) NOT NULL PRIMARY KEY,
    OrderDateTime DATETIME,
    PaymentApprovalNo VARCHAR(10),
    Status VARCHAR(50) DEFAULT 'Deliveried',
    CustomerID VARCHAR(10),
    StaffID VARCHAR(10),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID) ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (StaffID) REFERENCES InStoreStaff(StaffID) ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (PaymentApprovalNo) REFERENCES OrderPayRecord(PaymentApprovalNo) ON UPDATE CASCADE ON
DELETE CASCADE
);
go

-- Create the WalkInOrder table
CREATE TABLE WalkInOrder (
    OrderID VARCHAR(10) PRIMARY KEY,
    WalkInTime DATETIME,
    FOREIGN KEY (OrderID) REFERENCES OrderOnly(OrderID) ON UPDATE CASCADE ON DELETE NO ACTION
);
go

-- Create the PhoneOrder table
CREATE TABLE PhoneOrder (
    OrderID VARCHAR(10) PRIMARY KEY,
    PhoneStartTime DATETIME,
    PhoneEndTime DATETIME,
    FOREIGN KEY (OrderID) REFERENCES OrderOnly(OrderID) ON UPDATE CASCADE ON DELETE NO ACTION
);
go

-- Create the Delivery table
CREATE TABLE Delivery (
    OrderID VARCHAR(10) PRIMARY KEY,
    DeliveryAddress VARCHAR(100),
    DeliveryTime DATETIME,
    ShiftID VARCHAR(10),
    FOREIGN KEY (OrderID) REFERENCES PhoneOrder(OrderID) ON UPDATE CASCADE ON DELETE NO ACTION,
    FOREIGN KEY (ShiftID) REFERENCES DriverShift(ShiftID) ON UPDATE NO ACTION ON DELETE NO ACTION,
);
```

```sql
go

-- Create the Pickup table
CREATE TABLE Pickup (
    OrderID VARCHAR(10) PRIMARY KEY,
    PickupTime DATETIME,
    FOREIGN KEY (OrderID) REFERENCES PhoneOrder(OrderID) ON UPDATE CASCADE ON DELETE CASCADE
);
go

-- Create the QMenuOrder table
CREATE TABLE QMenuOrder (
    OrderID VARCHAR(10),
    ItemCode VARCHAR(10),
    Quantity INT,
    PRIMARY KEY (OrderID, ItemCode),
    FOREIGN KEY (OrderID) REFERENCES OrderOnly(OrderID) ON UPDATE CASCADE ON DELETE NO ACTION,
    FOREIGN KEY (ItemCode) REFERENCES MenuItem(ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION
);
go

-- Create the QMenuIngredient table
CREATE TABLE QMenuIngredient (
    ItemCode VARCHAR(10),
    IngredientCode VARCHAR(10),
    Quantity INT,
    PRIMARY KEY (ItemCode, IngredientCode),
    FOREIGN KEY (ItemCode) REFERENCES MenuItem(ItemCode) ON UPDATE CASCADE ON DELETE NO ACTION,
    FOREIGN KEY (IngredientCode) REFERENCES Ingredient(IngredientCode) ON UPDATE CASCADE ON DELETE NO
ACTION
);
go

-- Create the QIngredientIngredientOrder table
CREATE TABLE QIngredientIngredientOrder (
    IngredientCode VARCHAR(10),
    IngredientOrderNo VARCHAR(10),
    Price DECIMAL(10, 2),
    Quantity INT,
    PRIMARY KEY (IngredientCode, IngredientOrderNo),
    FOREIGN KEY (IngredientCode) REFERENCES Ingredient(IngredientCode) ON UPDATE CASCADE ON DELETE NO
ACTION,
    FOREIGN KEY (IngredientOrderNo) REFERENCES IngredientOrder(IngredientOrderNo) ON UPDATE CASCADE
ON DELETE NO ACTION,
);
go
```

Part 7: SQL Statements

```
--Input data of at least five rows for every table
--INSERT INTO table VALUES('','','','','');
--Insert data into MenuItem table
INSERT INTO MenuItem VALUES ('MENU001', 'Ultimate Cheese Lover''s','14 inch','Large Pizza', '23.95');
INSERT INTO MenuItem VALUES ('MENU002','Supreme','12 inch','Medium Pizza','19.85');
INSERT INTO MenuItem VALUES ('MENU003','Hawaiian','12 inch','Medium Pizza','18.95');
INSERT INTO MenuItem VALUES ('MENU004','MeatLover''s','16 inch','Extra Large pizza','25.95');
INSERT INTO MenuItem VALUES ('MENU005','BBQ Chicken','10 inch','Small pizza','15.95');
INSERT INTO MenuItem VALUES ('MENU006','Pepperoni Lover''s','14 inch','Large Pizza','22.95');
INSERT INTO MenuItem VALUES ('MENU007','Veggie Lover''s','14 inch','Large Pizza','22.95');
go

--Insert data into Ingredient table
INSERT INTO Ingredient VALUES ('IN001','Garlic','Vegetable','Fresh','7.5kg','10kg','3kg');
INSERT INTO Ingredient VALUES ('IN002','Mushroom','Vegetable','Fresh','10kg','8kg','3kg');
INSERT INTO Ingredient VALUES ('IN003','GreenBell Pepper','Vegetable','Fresh','10kg','7.5kg','3kg');
INSERT INTO Ingredient VALUES ('IN004','Onion','Vegetable','Fresh','10kg','8kg','3kg');
INSERT INTO Ingredient VALUES ('IN005','Pineapple','Fruit','Frozen','10kg','5kg','3kg');
INSERT INTO Ingredient VALUES ('IN006','Chicken','Meat','Chicken Breast','10kg','6kg','3kg');
INSERT INTO Ingredient VALUES ('IN007','Pork','Meat','Pork Belly','15kg','10kg','3kg');
INSERT INTO Ingredient VALUES ('IN008','Sausage','Processed Meat','Beef
Sausage','10kg','10kg','3kg');
INSERT INTO Ingredient VALUES ('IN009','Ham','Processed Meat','Champagne Leg
Ham','15kg','10kg','3kg')
INSERT INTO Ingredient VALUES ('IN010','Pepperoni','Processed Meat','Small
pepperoni','10kg','5kg','3kg');
INSERT INTO Ingredient VALUES ('IN011','Cheese','Contain Dairy','Mozzarella','10kg','8kg','3kg');
INSERT INTO Ingredient VALUES ('IN012','Italian Sauce','Pizza Sauce','Bazil & Oregano
sause','10kg','7kg','3kg');
INSERT INTO Ingredient VALUES ('IN013','Creamy Garlic Sauce','Pizza Sauce','Contains
Dairy','10kg','9kg','3kg');
INSERT INTO Ingredient VALUES ('IN014','Pizza Dough','Pizza Base','Gluten Free','20kg','10kg','3kg');
go

----Insert data into Supplier table
INSERT INTO Supplier VALUES ('SUPP1001','MaryButchers','0123123123','Hunter
Street','Newcastle','2300', 'marybutchers@gmail.com');
INSERT INTO Supplier VALUES ('SUPP1002','BidFood','0400444222','King Street','Newcastle','2300',
'bidfood@gmail.com');
INSERT INTO Supplier VALUES ('SUPP2000','FoodBomb','0291988452','Glebe Roas','Newcastle','2308',
'foodbomb@gmail.com');
INSERT INTO Supplier VALUES ('SUPP2001','FoodnDairy','023005445','Bull STreet','Newcastle','2300',
'foodndairy@gmail.com');
INSERT INTO Supplier VALUES ('SUPP2002','Bibica','0562125621','Tydlad Street','Newcastle','2299',
'bibica@gmail.com');
INSERT INTO Supplier VALUES ('SUPP2003','HunterFood','045612896','Tyrrell Street','Newcastle','2300',
'hunterfood@gmail.com');
INSERT INTO Supplier VALUES ('SUPP2004','MocoFood','0453289562','Union Street','Newcastle','2300',
'mocomoco@gmail.com');
go

--Insert data into IngredientOrder table
INSERT INTO IngredientOrder VALUES ('INOR1100', 129.90, '2023-09-10', '2023-09-20', 'Delivered', 'On
time delivery', 'SUPP1001');
INSERT INTO IngredientOrder VALUES ('INOR1200', 150.00, '2023-08-10', '2023-08-25', 'Delivered',
null, 'SUPP1002');
INSERT INTO IngredientOrder VALUES ('INOR1300', 170.25, '2023-10-01', '2023-10-15', 'Delivered',
null, 'SUPP1001');
```

```sql
INSERT INTO IngredientOrder VALUES ('INOR2100',  80.90, '2023-10-08', '2023-10-16', 'Delivered',
'Slightly late', 'SUPP2004');
INSERT INTO IngredientOrder VALUES ('INOR2200', 145.50, '2023-10-15', '2023-10-25', 'Delivered',
'Deliveried late', 'SUPP2002');
INSERT INTO IngredientOrder VALUES ('INOR2300', 120.75, '2023-10-10', '2023-10-24', 'In Progress',
null, 'SUPP2002');
go


-- Insert data into the Customer table
INSERT INTO Customer VALUES ('CUST001', 'Mary', 'Smith', '123 Hunter Street', 'Newcastle', '2300');
INSERT INTO Customer VALUES ('CUST002', 'Nick', 'Lomma', '456 Elm Avenue', 'Sydney', '2000');
INSERT INTO Customer VALUES ('CUST003', 'Paul', 'Brown', '789 Oak Road', 'Melbourne', '3000');
INSERT INTO Customer VALUES ('CUST004', 'Lisa', 'Chou', '101 Pine Lane', 'Brisbane', '4220');
INSERT INTO Customer VALUES ('CUST005', 'Michael', 'Lee', '222 Cedar Street', 'Perth', '6203');
go

-- Insert data into the CPhone table
INSERT INTO CPhone VALUES ('0423589456', 'CUST001');
INSERT INTO CPhone VALUES ('0123568459', 'CUST002');
INSERT INTO CPhone VALUES ('0125126563', 'CUST003');
INSERT INTO CPhone VALUES ('0148623259', 'CUST004');
INSERT INTO CPhone VALUES ('0259245621', 'CUST005');
go

-- Insert data into the DriverPay table
INSERT INTO DriverPay VALUES ('DP001', 2050.00, 460.00, 1520.00, '2023-08-10', '2023-08-01', '2023-
08-25', 120.00, 1590.00);
INSERT INTO DriverPay VALUES ('DP002', 1550.00, 360.00, 1460.00, '2023-10-15', '2023-10-01', '2023-
10-30', 90.00, 1450.00);
INSERT INTO DriverPay VALUES ('DP003', 1250.00, 450.00, 1860.00, '2023-09-17', '2023-08-15', '2023-
09-25', 100.00, 1630.00);
INSERT INTO DriverPay VALUES ('DP004', 1990.00, 390.00, 1520.00, '2023-06-12', '2023-05-26', '2023-
06-20', 195.00, 1425.00);
INSERT INTO DriverPay VALUES ('DP005', 2250.00, 520.00, 1580.00, '2023-11-02', '2023-10-16', '2023-
11-15', 80.00, 1675.00);
go

-- Insert data into the InStorePay table
INSERT INTO InStorePay VALUES ('IP001', 2510.00, 300.00, 1625.00, '2023-08-12', '2023-08-03', '2023-
08-15', 20.00, 80);
INSERT INTO InStorePay VALUES ('IP002', 1406.00, 253.00, 1230.00, '2023-10-12', '2023-10-01', '2023-
10-15', 19.00, 70);
INSERT INTO InStorePay VALUES ('IP003', 1600.00, 452.00, 1280.00, '2023-09-10', '2023-08-25', '2023-
09-15', 20.25, 67);
INSERT INTO InStorePay VALUES ('IP004', 1550.00, 125.00, 1240.00, '2023-07-12', '2023-06-26', '2023-
07-31', 18.75, 74);
INSERT INTO InStorePay VALUES ('IP005', 1452.00, 450.00, 1500.00, '2023-11-12', '2023-10-16', '2023-
11-30', 17.50, 74);
go

-- Insert data into the BankDetails table
INSERT INTO BankDetails VALUES ('123-456', 'John Smith');
INSERT INTO BankDetails VALUES ('789-012', 'Alice Johnson');
INSERT INTO BankDetails VALUES ('345-678', 'David Brown');
INSERT INTO BankDetails VALUES ('901-234', 'Sarah Davis');
INSERT INTO BankDetails VALUES('567-890', 'Michael Wilson');
go
-- Insert data into the DriverStaff table
```

```sql
INSERT INTO DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, Description, DriverLicNo, RatePerDelivery) VALUES ('DS001', 'John', 'Smith', '150 Cinre Street', 'Brisbane', '5962', '2632785', '123-456', '1234567890', 'Part-time', 'DL751023', 5.50);
INSERT INTO DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, Description, DriverLicNo, RatePerDelivery) VALUES ('DS002', 'Alice', 'Johnson', '89A Inue Avenue', 'Sydney', '2000', '245985230', '789-012', '9876543210', 'Part-time', 'DL752015', 4.00);
INSERT INTO DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, DriverLicNo, RatePerDelivery) VALUES ('DS003', 'David', 'Brown', '14B Danfin Street', 'Newcastle', '3000', '562123148', '345-678', '5432167890', 'DL423658', 4.25);
INSERT INTO DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, DriverLicNo, RatePerDelivery) VALUES ('DS004', 'Sarah', 'Davis', '52 Gine Street', 'Sydney', '4000', '245698745', '901-234', '2109876540', 'DL210987', 5.75);
INSERT INTO DriverStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, DriverLicNo, RatePerDelivery) VALUES ('DS005', 'Michael', 'Wilson', '33 Huneit Road', 'Adelaide', '6000', '125623456', '567-890', '6543217890', 'DL654321', 4.50);
go

-- Insert data into the InStoreStaff table
INSERT INTO InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, HourlyRate) VALUES ('ISS001', 'John', 'Smith', '123 Dervin Street', 'Newcastle', '2300', '123456789', '123-456', '1234567890', 20.00);
INSERT INTO InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, HourlyRate, Description) VALUES ('ISS002', 'Alice', 'Johnson', '456 Elm Street', 'Sydney', '2000', '987654321', '789-012', '9876543210', 18.50, 'Part-time');
INSERT INTO InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, HourlyRate) VALUES ('ISS003', 'David', 'Brown', '74 Oak Road', 'Melbourne', '3000', '543216789', '345-678', '5432167890', 21.00);
INSERT INTO InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, HourlyRate, Description) VALUES ('ISS004', 'Sarah', 'Davis', '101 Inien Lane', 'Brisbane', '4000', '210987654', '901-234', '2109876540', 19.75,'Part-time');
INSERT INTO InStoreStaff (StaffID, FirstName, LastName, Street, City, Postcode, TaxFileNo, BSB, AccNo, HourlyRate) VALUES ('ISS005', 'Michael', 'Wilson', '52 Aunit Street', 'Perth', '6000', '654321789', '567-890', '6543217890', 22.50);
go

-- Insert data into the DriverShift table
INSERT INTO DriverShift VALUES ('DSH001', 'DS001', 'DP001', '2023-10-12 08:00:00', '2023-10-12 12:00:00', 13);
INSERT INTO DriverShift VALUES ('DSH002', 'DS002', 'DP002', '2023-10-12 11:30:00', '2023-10-12 15:30:00', 17);
INSERT INTO DriverShift VALUES ('DSH003', 'DS003', 'DP003', '2023-10-12 08:30:00', '2023-10-12 14:50:00', 7);
INSERT INTO DriverShift VALUES ('DSH004', 'DS004', 'DP004', '2023-10-12 13:00:00', '2023-10-12 17:00:00', 10);
INSERT INTO DriverShift VALUES ('DSH005', 'DS005', 'DP005', '2023-10-12 10:00:00', '2023-10-12 15:30:00', 14);
go

-- Insert data into the InStoreShift table
INSERT INTO InStoreShift VALUES ('ISHI001', 'ISS001', 'IP001', '2023-10-12 10:00:00', '2023-10-12 18:00:00', 8);
INSERT INTO InStoreShift VALUES ('ISHI002', 'ISS002', 'IP002', '2023-10-11 10:30:00', '2023-10-12 15:30:00', 7);
INSERT INTO InStoreShift VALUES ('ISHI003', 'ISS003', 'IP003', '2023-10-11 09:30:00', '2023-10-12 17:30:00', 8);
INSERT INTO InStoreShift VALUES ('ISHI004', 'ISS004', 'IP004', '2023-10-12 13:00:00', '2023-10-12 17:00:00', 4);
INSERT INTO InStoreShift VALUES ('ISHI005', 'ISS005', 'IP005', '2023-10-12 11:00:00', '2023-10-12 18:00:00', 7);
go
```

```sql
-- Insert data into the OrderPayRecord table
INSERT INTO OrderPayRecord VALUES ('PAY001', 'Cash', 50.75);
INSERT INTO OrderPayRecord VALUES ('PAY002', 'Cash', 63.90);
INSERT INTO OrderPayRecord VALUES ('PAY003', 'Credit Card', 50.25);
INSERT INTO OrderPayRecord VALUES ('PAY004', 'Credit Card', 59.00);
INSERT INTO OrderPayRecord VALUES ('PAY005', 'Credit Card', 80.95);
INSERT INTO OrderPayRecord VALUES ('PAY006', 'Cash', 40.75);
INSERT INTO OrderPayRecord VALUES ('PAY007', 'Cash', 50.00);
INSERT INTO OrderPayRecord VALUES ('PAY008', 'Debit Card', 80.25);
INSERT INTO OrderPayRecord VALUES ('PAY009', 'Cash', 50.00);
INSERT INTO OrderPayRecord VALUES ('PAY010', 'Cash', 70.05);
INSERT INTO OrderPayRecord VALUES ('PAY011', 'Credit Card', 69.55);
INSERT INTO OrderPayRecord VALUES ('PAY012', 'Cash', 60.65);
INSERT INTO OrderPayRecord VALUES ('PAY013', 'Debit Card', 54.75);
INSERT INTO OrderPayRecord VALUES ('PAY014', 'Debit Card', 35.50);
INSERT INTO OrderPayRecord VALUES ('PAY015', 'Credit Card', 70.55);
go
-- Insert data into the OrderOnly table
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD001', '2023-10-12 12:05:00', 'PAY001', 'Processing', 'CUST001', 'ISS001');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD002', '2023-10-13 20:00:00', 'PAY002', 'CUST002', 'ISS002');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD003', '2023-10-13 12:11:00', 'PAY003', 'Processing', 'CUST003', 'ISS003');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD004', '2023-10-13 11:30:00', 'PAY004', 'CUST004', 'ISS004');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD005', '2023-10-14 19:30:00', 'PAY005', 'Processing', 'CUST005', 'ISS005');

INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD006', '2023-10-13 10:42:00', 'PAY006', 'Processing', 'CUST001', 'ISS001');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD007', '2023-10-12 14:41:00', 'PAY007', 'CUST002', 'ISS002');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD008', '2023-10-14 11:00:00', 'PAY008', 'Processing', 'CUST003', 'ISS003');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD009', '2023-10-14 13:25:00', 'PAY009', 'CUST004', 'ISS004');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD010', '2023-10-15 19:30:00', 'PAY010', 'Processing', 'CUST005', 'ISS005');

INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD0011', '2023-10-12 11:45:00', 'PAY011', 'CUST001', 'ISS001');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD0012', '2023-10-13 15:40:00', 'PAY012', 'CUST002', 'ISS002');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID) VALUES
('ORD0013', '2023-10-13 12:01:00', 'PAY013', 'Processing', 'CUST003', 'ISS003');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, CustomerID, StaffID) VALUES
('ORD0014', '2023-10-13 14:30:00', 'PAY014', 'CUST004', 'ISS004');
INSERT INTO OrderOnly (OrderID, OrderDateTime, PaymentApprovalNo, Status, CustomerID, StaffID)VALUES
('ORD0015', '2023-10-14 20:35:00', 'PAY015', 'Processing', 'CUST005', 'ISS005');
go

-- Insert data into the WalkInOrder table
INSERT INTO WalkInOrder VALUES ('ORD001', '2023-10-12 11:55:00');
INSERT INTO WalkInOrder VALUES ('ORD002', '2023-10-13 19:55:00');
INSERT INTO WalkInOrder VALUES ('ORD003', '2023-10-13 12:05:00');
INSERT INTO WalkInOrder VALUES ('ORD004', '2023-10-13 11:20:00');
INSERT INTO WalkInOrder VALUES ('ORD005', '2023-10-14 19:27:00');
go

-- Insert data into the PhoneOrder table
```

```sql
INSERT INTO PhoneOrder VALUES ('ORD006', '2023-10-13 10:38:00', '2023-10-13 10:45:00');
INSERT INTO PhoneOrder VALUES ('ORD007', '2023-10-12 14:35:00', '2023-10-12 14:45:00');
INSERT INTO PhoneOrder VALUES ('ORD008', '2023-10-14 10:45:00', '2023-10-14 11:00:00');
INSERT INTO PhoneOrder VALUES ('ORD009', '2023-10-14 13:15:00', '2023-10-14 13:28:00');
INSERT INTO PhoneOrder VALUES ('ORD010', '2023-10-15 19:20:00', '2023-10-15 19:35:00');

INSERT INTO PhoneOrder VALUES ('ORD0011', '2023-10-13 11:38:00', '2023-10-13 11:45:00');
INSERT INTO PhoneOrder VALUES ('ORD0012', '2023-10-12 15:35:00', '2023-10-12 15:45:00');
INSERT INTO PhoneOrder VALUES ('ORD0013', '2023-10-14 11:45:00', '2023-10-14 12:00:00');
INSERT INTO PhoneOrder VALUES ('ORD0014', '2023-10-14 14:15:00', '2023-10-14 14:28:00');
INSERT INTO PhoneOrder VALUES ('ORD0015', '2023-10-15 20:20:00', '2023-10-15 20:35:00');
go

-- Insert data into the Delivery table
INSERT INTO Delivery VALUES ('ORD0011', '123 Elm Street, Newcastle', '2023-10-15 12:30:00',
'DSH001');
INSERT INTO Delivery VALUES ('ORD0012', '456 Oak Road, Newcastle', '2023-10-16 16:00:00', 'DSH002');
INSERT INTO Delivery VALUES ('ORD0013', '789 Pine Lane, Newcastle', '2023-10-17 12:20:00', 'DSH003');
INSERT INTO Delivery VALUES ('ORD0014', '101 Cedar Street, Newcastle', '2023-10-18 15:00:00',
'DSH004');
INSERT INTO Delivery VALUES ('ORD0015', '222 Birch Avenue, Newcastle', '2023-10-19 20:50:00',
'DSH005');
go

-- Insert data into the Pickup table
INSERT INTO Pickup VALUES ('ORD006', '2023-10-13 11:10:00');
INSERT INTO Pickup VALUES ('ORD007', '2023-10-12 15:00:00');
INSERT INTO Pickup VALUES ('ORD008', '2023-10-14 11:45:00');
INSERT INTO Pickup VALUES ('ORD009', '2023-10-14 14:30:00');
INSERT INTO Pickup VALUES ('ORD010', '2023-10-15 20:05:00');
go
-- Insert data into the QMenuOrder table
INSERT INTO QMenuOrder VALUES ('ORD001', 'MENU007', 1);
INSERT INTO QMenuOrder VALUES ('ORD001', 'MENU002', 1);
INSERT INTO QMenuOrder VALUES ('ORD002', 'MENU003', 1);
INSERT INTO QMenuOrder VALUES ('ORD002', 'MENU006', 1);
INSERT INTO QMenuOrder VALUES ('ORD002', 'MENU005', 1);
INSERT INTO QMenuOrder VALUES ('ORD003', 'MENU006', 1);
INSERT INTO QMenuOrder VALUES ('ORD004', 'MENU005', 3);
INSERT INTO QMenuOrder VALUES ('ORD005', 'MENU002', 1);
INSERT INTO QMenuOrder VALUES ('ORD006', 'MENU004', 2);
INSERT INTO QMenuOrder VALUES ('ORD007', 'MENU003', 4);
INSERT INTO QMenuOrder VALUES ('ORD008', 'MENU002', 1);
INSERT INTO QMenuOrder VALUES ('ORD008', 'MENU001', 1);
INSERT INTO QMenuOrder VALUES ('ORD009', 'MENU002', 3);
INSERT INTO QMenuOrder VALUES ('ORD010', 'MENU004', 3);
INSERT INTO QMenuOrder VALUES ('ORD0011', 'MENU004', 2);
INSERT INTO QMenuOrder VALUES ('ORD0012', 'MENU004', 1);
INSERT INTO QMenuOrder VALUES ('ORD0013', 'MENU003', 2);
INSERT INTO QMenuOrder VALUES ('ORD0013', 'MENU001', 1);
INSERT INTO QMenuOrder VALUES ('ORD0014', 'MENU003', 2);
INSERT INTO QMenuOrder VALUES ('ORD0015', 'MENU005', 1);
INSERT INTO QMenuOrder VALUES ('ORD0015', 'MENU006', 1);
INSERT INTO QMenuOrder VALUES ('ORD0015', 'MENU001', 1);
go

-- Insert data into the QMenuIngredient table
INSERT INTO QMenuIngredient VALUES ('MENU001', 'IN001', 2);
INSERT INTO QMenuIngredient VALUES ('MENU001', 'IN011', 1);
INSERT INTO QMenuIngredient VALUES ('MENU002', 'IN002', 2);
INSERT INTO QMenuIngredient VALUES ('MENU002', 'IN003', 1);
```

```sql
INSERT INTO QMenuIngredient VALUES ('MENU002', 'IN012', 4);
INSERT INTO QMenuIngredient VALUES ('MENU003', 'IN014', 1);
INSERT INTO QMenuIngredient VALUES ('MENU003', 'IN005', 2);
INSERT INTO QMenuIngredient VALUES ('MENU003', 'IN012', 2);
INSERT INTO QMenuIngredient VALUES ('MENU004', 'IN006', 3);
INSERT INTO QMenuIngredient VALUES ('MENU004', 'IN010', 2);
INSERT INTO QMenuIngredient VALUES ('MENU004', 'IN014', 5);
INSERT INTO QMenuIngredient VALUES ('MENU005', 'IN006', 2);
INSERT INTO QMenuIngredient VALUES ('MENU005', 'IN005', 1);
INSERT INTO QMenuIngredient VALUES ('MENU006', 'IN005', 3);
INSERT INTO QMenuIngredient VALUES ('MENU007', 'IN010', 2);
INSERT INTO QMenuIngredient VALUES ('MENU007', 'IN013', 1);
go
-- Insert data into the QIngredientIngredientOrder table
INSERT INTO QIngredientIngredientOrder VALUES ('IN001', 'INOR1100', 12.50, 5);
INSERT INTO QIngredientIngredientOrder VALUES ('IN002', 'INOR1100',  8.75, 3);
INSERT INTO QIngredientIngredientOrder VALUES ('IN003', 'INOR1200',  6.20, 4);
INSERT INTO QIngredientIngredientOrder VALUES ('IN004', 'INOR1200',  9.10, 2);
INSERT INTO QIngredientIngredientOrder VALUES ('IN005', 'INOR1300', 14.00, 6);
INSERT INTO QIngredientIngredientOrder VALUES ('IN006', 'INOR2100',  7.50, 4);
INSERT INTO QIngredientIngredientOrder VALUES ('IN007', 'INOR2100', 10.25, 3);
INSERT INTO QIngredientIngredientOrder VALUES ('IN008', 'INOR2200',  5.60, 5);
INSERT INTO QIngredientIngredientOrder VALUES ('IN009', 'INOR2200', 12.75, 2);
INSERT INTO QIngredientIngredientOrder VALUES ('IN010', 'INOR2300',  8.30, 4);
go

SELECT * FROM MenuItem
SELECT * FROM Ingredient
SELECT * FROM Supplier
SELECT * FROM IngredientOrder
SELECT * FROM Customer
SELECT * FROM CPhone
SELECT * FROM DriverPay
SELECT * FROM InStorePay
SELECT * FROM BankDetails
SELECT * FROM DriverStaff
SELECT * FROM InStoreStaff
SELECT * FROM DriverShift
SELECT * FROM InstoreShift
SELECT * FROM OrderPayrecord
SELECT * FROM OrderOnly
SELECT * FROM WalkInOrder
SELECT * FROM PhoneOrder
SELECT * FROM Delivery
SELECT * FROM Pickup
SELECT * FROM QMenuOrder
SELECT * FROM QMenuIngredient
SELECT * FROM QIngredientIngredientOrder
go

-- Implement Queries
-- Q.1 For an instore staff with id number ISS001,
-- print his/her first name, last name, and hourly payment rate.
SELECT FirstName, LastName, HourlyRate
FROM InStoreStaff
WHERE StaffID = 'ISS001';


-- Q.2 List all the shift details of a delivery staff with first name JOHN and last name SMITH
-- between date 2023-10-12 and 2023-10-20
```

```sql
SELECT SH.ShiftID, SH.StaffID, SH.RecordID, SH.StartDateTime, SH.EndDateTime, SH.DeliveriesNum
FROM DriverStaff s
JOIN DriverShift SH
ON sh.staffID = s.staffID
WHERE S.FirstName = 'John' AND S.LastName = 'Smith'
AND SH.StartDateTime >= '2023-10-12 00:00:00'
AND SH.EndDateTime <=  '2023-10-20 23:59:59';

-- Q.3 List all the order details of the orders that are made by a walk-in customer
-- with first name Lisa and last name Chou between date 2023-10-10 and 2023-10-20.
SELECT W.WalkInTime, O.OrderDateTime, O.Status, P.PaymentMethod, P.TotalAmountDue
FROM OrderOnly O
JOIN WalkInOrder W
ON W.OrderID = O.OrderID
JOIN Customer C
ON C.CustomerID = O.CustomerID
JOIN OrderPayRecord P
ON P.PaymentApprovalNo = O.PaymentApprovalNo
WHERE C.FirstName ='Lisa' AND C.LastName = 'Chou'
AND O.OrderDateTime BETWEEN '2023-10-10 00:00:00' AND '2023-10-20 23:59:59'

-- Q.4 List the names of the menu items that are ordered in the current year. Note the
-- current year is the current year that is decided by the system.
SELECT DISTINCT M.Name
FROM MenuItem M
JOIN QMenuOrder Q ON M.ItemCode = Q.ItemCode
JOIN OrderOnly O ON O.OrderID = Q.OrderID
WHERE YEAR(O.OrderDateTime) = YEAR(CURRENT_TIMESTAMP)


-- Q.5 List the name(s) of the ingredient(s) that was/were supplied by the supplier
-- with supplier ID SUPP1001 on date 2023-09-20
SELECT DISTINCT I.Name
FROM Ingredient I
JOIN QIngredientIngredientOrder Q
ON Q.IngredientCode = I.IngredientCode
JOIN  IngredientOrder O
ON O.IngredientOrderNo = Q.IngredientOrderNo
WHERE O.SupplierID = 'SUPP1001'
AND O.DateSupplied = '2023-09-20';
```